

 MESH CORE SCALING ARCHITECTURE

MeshCore Scaling Strategy

Region Scope & Channel use for Middle TN / NashMesh

PRESENTER



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EVENT



Wednesday, Aug 19, 2026 - NashMesh Meetup

⚠ The Problem: Flood Traffic & Airtime Saturation

- ❌ **Duty Cycle Collapse:** Unscoped messages (*) repeat across every node in RF reach, exhausting airtime.
- ❌ **Hidden Node Collisions:** High-elevation mountain relays hear competing nodes that can't hear each other.
- ❌ **The "Bot & Telemetry" Tax:** Automated pings flood wide-area backbones, blocking human emergency chat.
- ❌ **Non-Linear Scale Failure:** Unbounded mesh flooding fails as node density increases across city boundaries.

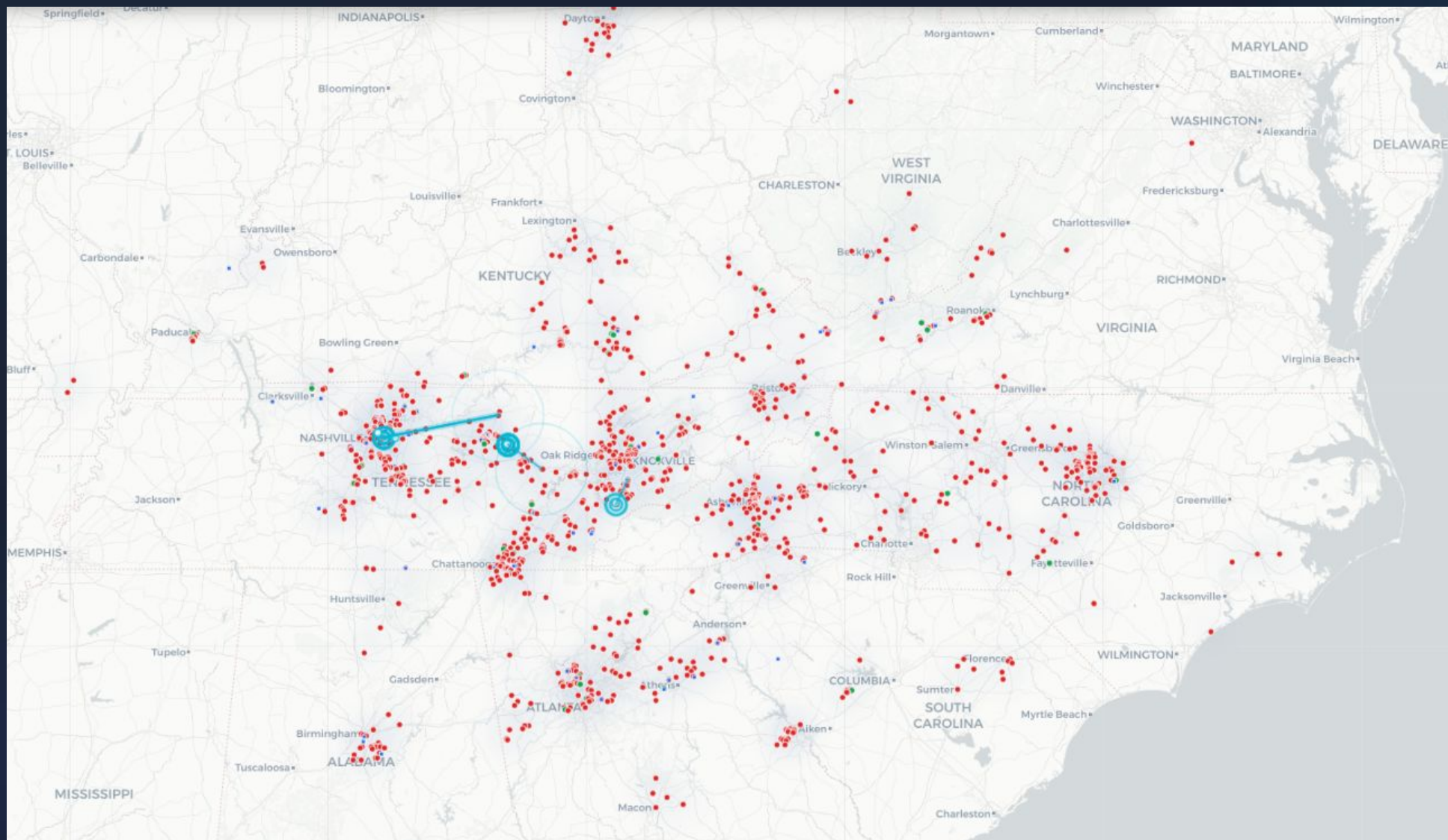
⚡ Unscoped Flooding (*)

📡 **Nashville Chat** → Floods TN → Leaks to GA, KY, NC, SC, AL, VA and beyond

🛡 Scoped Region (us-tn-bna)

📡 **Local BNA Chat** → Contained locally → Blocked at regional backbone

⚠ MeshCore US southeast: Multiple merged states



Architectural Distinction: Channels vs. Regions

Channels (1-Byte Hash)

"Who gets to read the message"

- ✓ Controls encryption & access group membership. Public vs *#nashmesh*
- ✓ Changing channel name isolates chat visibility.
- ⚠ **Crucial Limit:** Channels do NOT limit RF propagation! A local channel message still floods distant nodes if unscoped.

Regions (2-Byte Hash)

"Where the message gets forwarded"

- ✓ Controls repeater retransmission rules on RF layer.
- ✓ Region names compile into a fixed 2-byte transport hash over the air.
- ⚡ **Zero Payload Penalty:** Long region names like *us-tn-bna* consume 0 extra airtime bytes!

🏠 Region Deep Dive: Hierarchy & Propagation Rules

⬇️ Top-Down Cascading (Allowed)

Messages sent on parent region (e.g. us-tn) automatically deliver down to child repeaters (us-tn-mid, us-tn-bna). Great for statewide alerts!

🚫 Bottom-Up Containment (Blocked)

Local chatter sent on child region (us-tn-bna) is ignored by parent-only backbones (us-southeast). Protects wide-area airspace!



⚠️ High profile repeaters carry multiple regions to bridge interstate corridors cleanly.

Proposed Middle TN / NashMesh Region use

Scope Level	Region	Recommended Channels	Notes & Scope Alignment
Flood	*	Any/default	Unscoped flood across all repeaters unless denied
Country	us	Public, #bot, #test	Nationwide layer; handles wide area & cross-country.
Country Region	us-southeast	Public, #bot, #test	Multi-state corridor across TN, GA, NC, AL, KY, VA.
State	us-tn	#tn, #tenntalk	Statewide coordination across all Grand Divisions.
State Division	us-tn-mid	#tn-middle	Middle TN umbrella. Flanked by us-tn-east & us-tn-west.
IATA Metro	us-tn-bna	#bna-bot, #bna-wx, #bna-test, #wardriving	Nashville metro aligned with global MeshMapper.
Local / Tactical	nashmesh	#nashmesh	Dedicated local infrastructure or tactical ops.

Regional Alignment & Ecosystem Consensus



Grand Divisions

us-tn-mid mirrors us-tn-east (Knoxville) and us-tn-west (Memphis), allowing regional backbones to isolate terrain-bound chatter.



Interstate Corridors

High-elevation relays along I-65, I-75, and I-40 bind us-southeast alongside state regions for cross-border emergency bridging.



MeshMapper Standard

Standardizing on IATA codes (bna, cha, tys, atl) matches global node map conventions across companion apps.

☰ Best Practices vs. Anti-Patterns

✓ Best Practices

- ✓ **Standardize Client Defaults:** Set client app default scope to us-tn-bna for routine chat.
- ✓ **Stack High Profile Repeaters:** Add multiple regions (us-southeast, us-tn, us-tn-mid) on mountain relays.
- ✓ **Phased Wildcard Filter:** Gradually apply region denyf * as scope adoption matures. (maybe/eventually/never?)

✗ Anti-Patterns

- ✗ **Flat Unscoped Traffic:** Leaving repeaters on default open * wildcard floods all regional nodes.
- ✗ **Overly Deep Names:** Inventing hyper-long regions like us-tn-davidson-east increases typo risk.
- ✗ **Channel-Only Isolation:** Expecting `#local-chat` channel name to restrict RF flood distance.

> Repeater CLI Reference Implementation

MeshCore Serial Console – repeater_config.sh

1. Reset existing region table (optional)

region clear

2. Build top-down parent-child hierarchy

region put us

region put us-southeast us

region put us-tn us-southeast

region put us-tn-mid us-tn

region put us-tn-bna us-tn-mid

3. Add standalone local tactical scope & save to flash

region put nashmesh

region save

Chattanooga Mesh east TN region example

chattame.sh

03 · REGION SCOPING

Repeater Region Hierarchy

Chattanooga MeshCore region tree:
Only for human readability.
Your repeater does not care.



■ us-tn-east ■ us-tn-cha — Chattanooga

Repeater/testing regions currently active: * · us · us-southeast · us-tn · us-tn-east · us-tn-cha — all with Flood (F) allowed.



Region CLI by Firmware

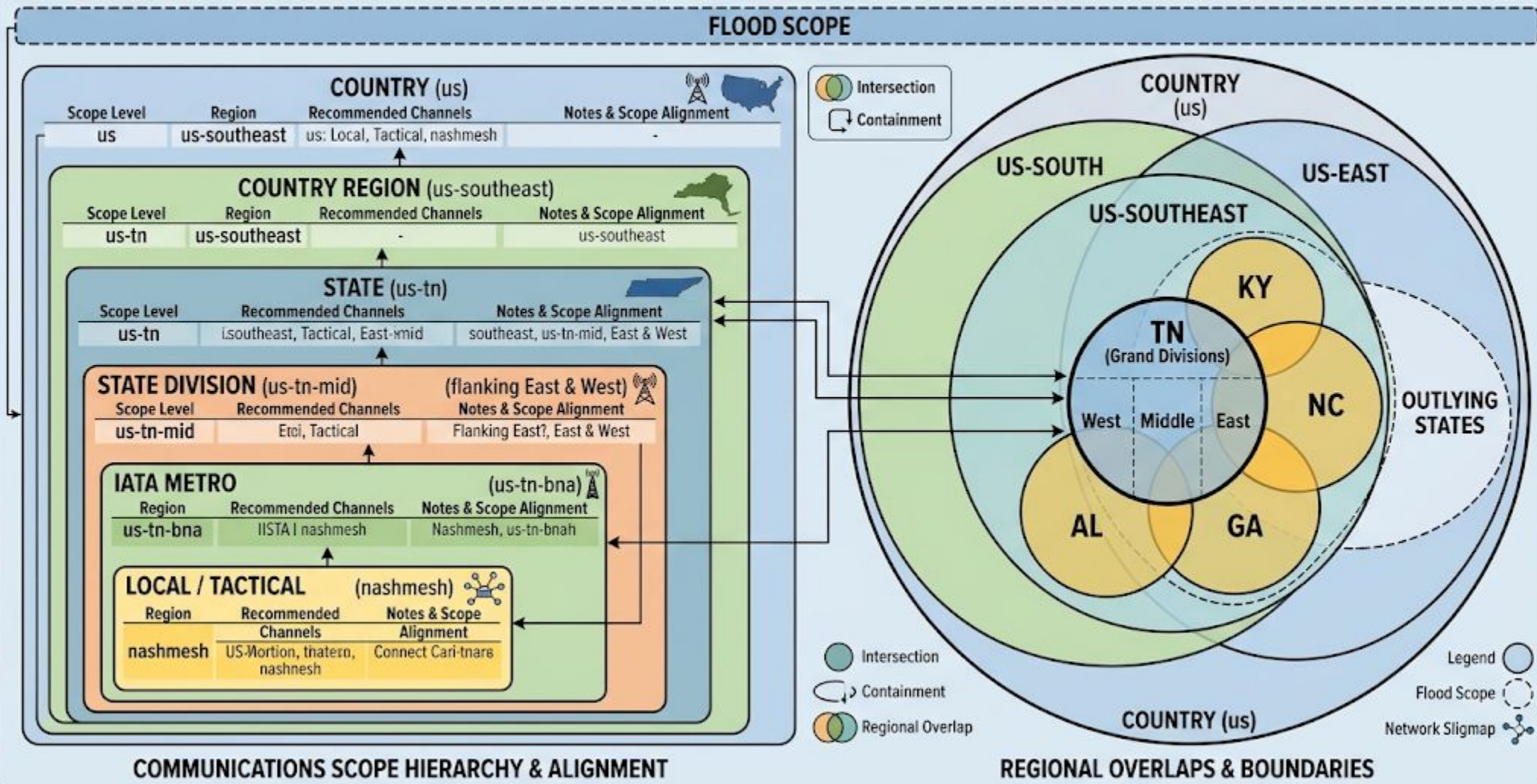
- v1.17
- v1.16
- v1.15 (untested)
- v1.14 (untested)

```
> region def us us-southeast us-tn us-tn-east|us-tn us-tn-cha
> region save
```

v1.17 defines the full hierarchy in a single command.

🤝 State of TN region scope overlap and hierarchy

STATE OF TN COMMUNICATION SCOPES: OVERLAP & HIERARCHY





Questions & Community Discussion

Let's coordinate on repeater region scope rollouts
across Middle Tennessee and surrounding meshes!

 **NashMesh** Community

|  Scope: `us-tn-bna`

|  Channel: `#bna-chat`

Image Sources



<https://media.istockphoto.com/id/1438972076/photo/mobile-phone-base-station-aerial-view.jpg?s=612x612&w=0&k=20&c=a5jKZHWMHcOA95YOfz8ZB9EHFqANtoMWG9eb-tTaHPg=>

Source: www.istockphoto.com